Define the Laplacian Operator as:

$$\nabla^2 = \sum_{i=1}^{n} \frac{\partial^2}{\partial x_i^2}$$

Play a role in the grayscale image:

$$\nabla^2 f(x, y) = \frac{\partial^2 f(x, y)}{\partial x^2} + \frac{\partial^2 f(x, y)}{\partial y^2}$$

According to the inverse Fourier transform:

$$f(x, y) = \mathcal{F}^{-1}\left[F(u, v)\right] = \iint_{-\infty}^{+\infty} F(u, v) \mathrm{e}^{j2\pi(ux+vy)} \mathrm{d}u \mathrm{d}v$$

Discrete form:

$$f(x, y) = \mathcal{F}^{-1}\left[F(u, v)\right] = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) \mathrm{e}^{j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

First-order partial derivative:

$$\frac{\partial f(x, y)}{\partial x} = j2\pi u \iint_{-\infty}^{+\infty} F(u, v) \mathrm{e}^{j2\pi(ux+vy)} \mathrm{d}u \mathrm{d}v$$

Second-order partial derivative:

$$\frac{\partial f^2(x, y)}{\partial x^2} = (j2\pi u)^2 \iint_{-\infty}^{+\infty} F(u, v) \mathrm{e}^{j2\pi(ux+vy)} \mathrm{d}u \mathrm{d}v$$

Therefore:

$$\mathcal{F}\left[\frac{\partial f^2(x, y)}{\partial x^2}\right] = -(2\pi u)^2 \mathcal{F}\left\{\mathcal{F}^{-1}\left[F(u, v)\right]\right\} = -4\pi^2 u^2 F(u, v)$$

In conclusion, the Fourier transform of the Laplacian operator is as:

$$\mathcal{F}\left[\nabla^2 f(x, y)\right] = -4\pi^2(u^2 + v^2) F(u, v)$$

Laplace operator convolution kernel $p(x, y)$ is as follows:

$$p(x, y) = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$